

Finite Sets

JISHNU GHOSH

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§1 Problems

§1.1 Some fundamental results

1. If $\mathcal{A}$ is a collection of distinct subsets of the n -set S such that $A_i \cap A_j = \emptyset$ for all $A_i, A_j \in \mathcal{A}$, then $|\mathcal{A}| \leq 2^{n-1}$. Further, if $|\mathcal{A}| < 2^{n-1}$, $\mathcal{A}$ can be extended to a collection of 2^{n-1} subsets also satisfying the given intersection property.
2. Prove that if $A_1, \dots, A_m$ are distinct subsets of an n -set S such that for each pair i, j , $A_i \cup A_j \neq S$, then $m \leq 2^{n-1}$.
3. [**Lubell-Yamamoto-Meshalkin Inequality**] Let $A_1, A_2, \dots, A_s$ be subsets of $\{1, 2, \dots, n\}$ such that A_i is not a subset of A_j for any i and j . Let a_i denote $|A_i|$ for each i . Show that

$$\sum_{i=1}^s \frac{1}{\binom{n}{a_i}} \leq 1.$$

4. [**Sperner's Theorem**] Let $A_1, A_2, \dots, A_s$ be subsets of $\{1, 2, \dots, n\}$ such that A_i is not a subset of A_j for any i and j . Such a family of sets is known as an *antichain*. Show that

$$s \leq \binom{n}{\lfloor n/2 \rfloor}.$$

5. [**Bollobás' Theorem**] Let $A_1, A_2, \dots, A_m, B_1, B_2, \dots, B_m$ be subsets of an n -set S such that $A_i \cap B_i = \emptyset$ if and only if $i = j$. Let $a_i = |A_i|$ and $b_i = |B_i|$. Then

$$\sum_{i=1}^m \frac{1}{\binom{a_i+b_i}{a_i}} \leq 1.$$

6. [**Griggs 1984**] Let $f(n, k)$ denote the largest value of m for which it is possible to find m chains of $k+1$ distinct subsets of an n -set S such that no member of any chain is a subset of a member of any other chain. Then

$$f(n, k) = \binom{n-k}{\lfloor (n-k)/2 \rfloor}.$$

§1.2 Applications of Sperner's theorem and Bollobás' theorem

1. Show that if $\mathcal{A}$ is an antichain of subsets of an n -set with $|A| \leq h \leq n/2$ for all $A \in \mathcal{A}$, then $|\mathcal{A}| \leq \binom{n}{h}$.
2. [**Erdős 1945**] Let $x_1, x_2, \dots, x_n$ be real numbers such that $|x_i| \geq 1$ for each i , and let J be any interval of length 2 open at atleast one end. Then the number of linear combinations $\sum_{i=1}^n \varepsilon_i x_i$ ($\varepsilon_i = \pm 1$) lying in J is at most $\binom{n}{\lfloor n/2 \rfloor}$.
3. (Alternate formulation of the above) Let $x_1, x_2, \dots, x_n$ be real numbers such that $|x_i| \geq 1$ for each i , and let J be any interval of length 1 open at atleast one end. Then the number of linear combinations $\sum_{i=1}^n \varepsilon_i x_i$ ($\varepsilon_i = 0$ or 1) lying in J is at most $\binom{n}{\lfloor n/2 \rfloor}$.
4. Let $A_1, \dots, A_m, B_1, \dots, B_m$ be subsets of an n -set S such that $A_i \subseteq B_j$ if and only if $i = j$. Show that

$$\sum_{i=1}^m \frac{1}{\binom{n-|B_i-A_i|}{|A_i|}} \leq 1.$$

5. [**Tuza's variant of Bollobás theorem**] Let $\mathcal{A} = \{A_1, \dots, A_m\}$ be a collection of subsets of an n -set S and let $a_i = |A_i|$. Call a subset B_i of A_i an *own-subset* of A_i if $B_i \not\subseteq A_j$ for all $j \neq i$. Prove that if every member of $\mathcal{A}$ has an own-subset, then

$$\sum_{i=1}^m \frac{1}{\binom{n-a_i-b_i}{b_i}} \leq 1$$

where $b_i = |B_i|$.

§1.3 Application of Sperner's theorem in Olympiads

1. (STEMS 2025 CS CAT B) The famous detective Sherlock Holmes was summoned on a murder case. There is a murderer in a village of 2025 peoples. Sherlock knew that a witness had seen what happened, but he had no other information about the killer or the witness. No one else has information and cannot help. Sherlock can call a group of people to his office for interrogation. If the witness is in that group and the murderer is not in that group, the witness will reveal who the killer is. Each such interrogation session is called a “meeting”. Determine the minimum number of meetings Sherlock Holmes can conduct to identify the murderer.